

Paper Figures (High-Resolution, 1200 dpi)

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0. Purpose

This script regenerates the final set of figures selected for the manuscript, exported at **1200 dpi** into a dedicated `results/paper_images` folder. It reuses the objects already computed by scripts `3_DifferentialExpression.Rmd`, `4_FunctionalCategorization.Rmd`, and `5_AdvancedAnalysis.Rmd` — run those first if the intermediate files listed below are missing.

Figures produced:

1. Volcano plot — all genes (Treatment vs Control)
2. KEGG pathway enrichment (GSEA ridgeplot, “Photosynthesis” pathways excluded)
3. Protein-protein interaction (PPI) network, high confidence (STRING score > 700)
4. Volcano plot — virulence-associated genes highlighted
5. Heatmap — virulence-associated genes
6. Volcano plot — transcriptional regulators highlighted

```
library(DESeq2)
library(EnhancedVolcano)
library(tidyverse)
library(pheatmap)
library(viridis)
library(ggrepel)
library(clusterProfiler)
library(enrichplot)
library(KEGGREST)
library(STRINGdb)
```

```

library(igraph)
library(ggraph)
library(tidygraph)
library(magick)
library(pdftools)

# Dedicated output folder for final paper figures (1200 dpi)
paper_dir <- "../results/paper_images"
dir.create(paper_dir, showWarnings = FALSE, recursive = TRUE)

DPI <- 1200

# Consistent condition color palette (used across all plots)
pal_control <- "#4575B4"
pal_treatment <- "#D73027"
cond_palette <- c("Control" = pal_control, "Treatment" = pal_treatment)
pal_diverging <- colorRampPalette(c("#4575B4", "white", "#D73027"))(100)

```

1. Load required intermediate data

```

# Full DESeq2 results table (all genes, script 3)
res <- read.csv("../results/deseq_results_Treatment_vs_Control.csv", row.names = 1)

# Strict DEG subset (script 3)
deg_table <- read_csv("../results/deg_Treatment_vs_Control.csv", show_col_types = FALSE) %>%
  rename(GeneID_full = ...1) %>%
  mutate(GeneID = str_extract(GeneID_full, "WP_[0-9]+\\.[0-9]+"))
all_deg_ids <- trimws(deg_table$GeneID)

# Full annotated results (all genes + eggNOG, script 4)
full_annotation <- read_csv("../results/advanced_analysis/full_annotated_results.csv",
  show_col_types = FALSE)

# eggNOG annotation (KEGG dictionary source)
eggnog_data <- read.delim("../data/annotation.tsv")

# Metadata and normalized counts (needed for the virulence heatmap)
metadatos <- read.csv(file = "../data/metadata_all.csv", row.names = 1)
load(file = "../results/dds_norm.RData")
normcounts <- counts(dds_norm, normalized = TRUE)
colnames(normcounts) <- metadatos$Replicates

# Gene-name label map (long ID -> short name, e.g. pelA)
label_map <- full_annotation %>%
  filter(!is.na(Preferred_name), !Preferred_name %in% c("NA", "-", "")) %>%
  dplyr::select(GeneID, Preferred_name) %>%
  distinct(GeneID, .keep_all = TRUE)

short_wp <- function(x) str_extract(x, "WP_[0-9]+\\.[0-9]+")

```

```

make_labels <- function(long_ids) {
  wp <- short_wp(long_ids)
  labels <- ifelse(wp %in% label_map$GeneID,
                  label_map$Preferred_name[match(wp, label_map$GeneID)],
                  wp)
  labels[is.na(labels)] <- wp[is.na(labels)]
  labels
}

# Virulence factors table (script 5_AdvancedAnalysis)
virulence_df <- read_csv("../results/advanced_analysis/virulence_factors_table.csv",
                        show_col_types = FALSE)
virulence_genes <- virulence_df$GeneID

```

2. Figure 1 — Volcano Plot: All Genes

```

row_labels <- make_labels(rownames(res))
sig_mask <- !is.na(res$padj) & res$padj < 0.01 & abs(res$log2FoldChange) > 1

# --- Mixed ranking: combines significance (padj) and magnitude (log2FC) ---
# Uses rank() instead of raw normalized values, since many genes share the
# same clamped padj (~0), which would otherwise distort a min-max score.
# Takes pmin(rank_padj, rank_lfc) instead of summing, so a gene qualifies
# if it stands out strongly on EITHER axis - this prevents the hundreds of
# padj-clamped genes from dominating the selection and crowding out genes
# that are extreme only in log2FC.
N_TOP_LABELS <- 20

rank_df <- res[sig_mask, ] %>%
  as.data.frame() %>%
  rownames_to_column("GeneID_full") %>%
  mutate(
    neglog10_padj = -log10(padj),
    abs_lfc       = abs(log2FoldChange),
    rank_padj     = rank(neglog10_padj, ties.method = "average"),
    rank_lfc      = rank(abs_lfc, ties.method = "average"),
    best_rank     = pmin(rank_padj, rank_lfc)
  ) %>%
  arrange(desc(best_rank)) %>%
  slice_head(n = N_TOP_LABELS)

# --- Extra rule: always label genes with a very large fold change ---
# Independent of the ranking above, any significant gene whose |log2FC|
# exceeds this threshold gets labeled too, so large-effect outliers are
# never dropped just for missing the top-N cutoff.
FC_LABEL_THRESHOLD <- 2

fc_extreme_ids <- res[sig_mask, ] %>%
  as.data.frame() %>%

```

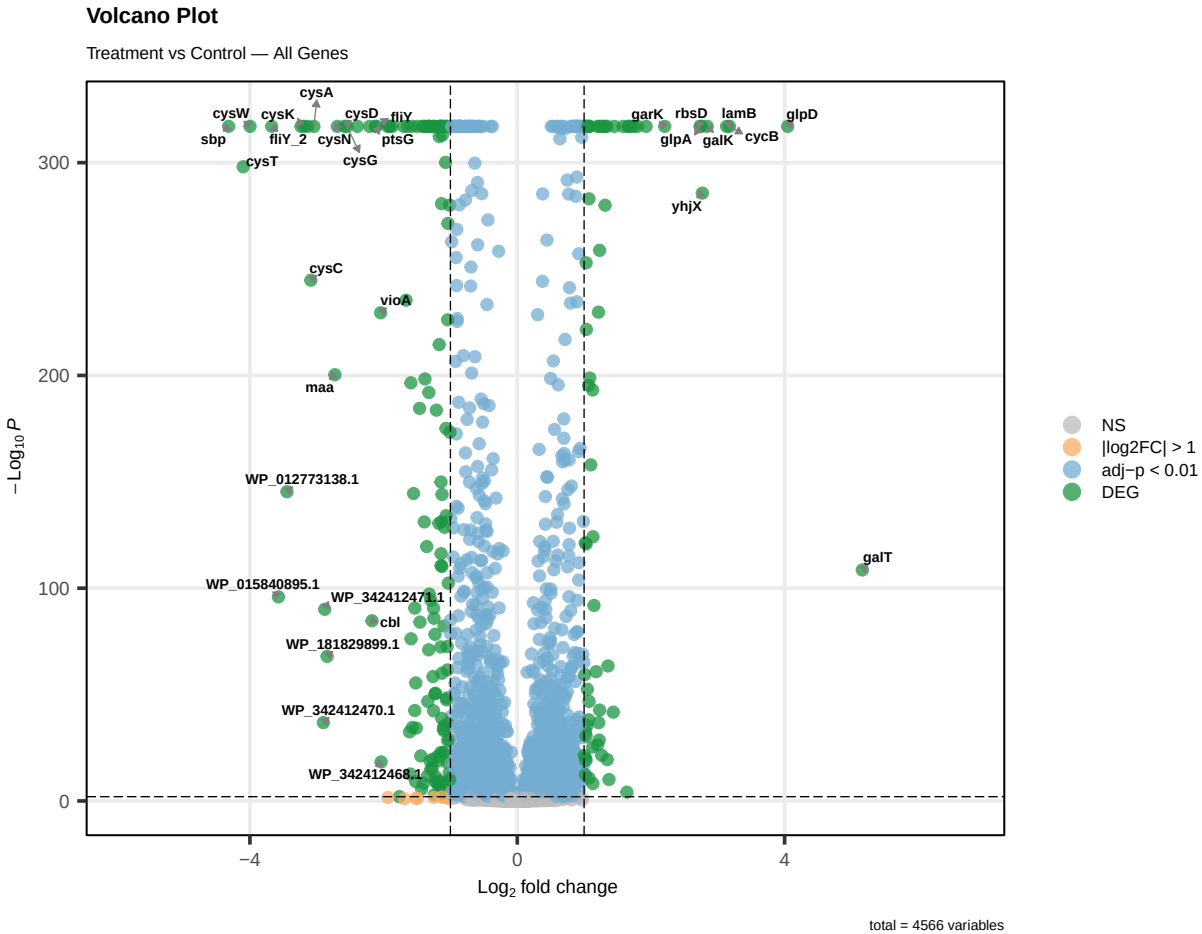
```

rownames_to_column("GeneID_full") %>%
filter(abs(log2FoldChange) > FC_LABEL_THRESHOLD) %>%
pull(GeneID_full)

# --- Build a position-matched label vector (avoids selectLab name-collision) ---
# selectLab in EnhancedVolcano matches by label TEXT, so if two different
# rows share the same short gene symbol (e.g. two entries both mapping to
# "cysG"), it labels every occurrence of that text - including
# non-significant duplicates elsewhere in the table. Instead, we blank out
# all labels except the exact selected rows, matched by GeneID (rowname).
top_ids <- union(rank_df$GeneID_full, fc_extreme_ids)
plot_labels <- ifelse(rownames(res) %in% top_ids, row_labels, "")

volc_all <- EnhancedVolcano(res,
  lab = plot_labels,
  selectLab = plot_labels[plot_labels != ""],
  x = "log2FoldChange",
  y = "padj",
  axisLabSize = 12,
  pCutoff = 0.01,
  FCcutoff = 1,
  col = c("#BDBDBD", "#FDAE61", "#74ADD1", "#1A9641"),
  colAlpha = 0.75,
  pointSize = 2.5,
  title = "Volcano Plot",
  titleLabSize = 14,
  subtitle = "Treatment vs Control - All Genes",
  subtitleLabSize = 11,
  captionLabSize = 9,
  labSize = 3.2,
  labFace = "bold",
  labCol = "black",
  legendLabels = c("NS", "|log2FC| > 1", "adj-p < 0.01", "DEG"),
  legendPosition = "right",
  legendLabSize = 11,
  legendIconSize = 4,
  drawConnectors = TRUE,
  widthConnectors = 0.4,
  colConnectors = "grey50",
  border = "full",
  borderWidth = 0.6,
  gridlines.major = TRUE,
  gridlines.minor = FALSE)
volc_all

```



```
ggsave(filename = file.path(paper_dir, "Fig1_Volcano_AllGenes.tiff"),
        plot = volc_all, width = 11, height = 8.5, dpi = DPI, compression = "lzw")
```

3. Figure 2 — KEGG Pathway Enrichment (GSEA Ridgeplot)

We run a Gene Set Enrichment Analysis (GSEA) over the full ranked gene list (by log2FoldChange) against KEGG pathways, then plot the result as a ridgeplot. The “Photosynthesis” pathway(s) are excluded from the plot, since they are not biologically relevant for this comparison.

```
term2gene_kegg <- eggnog_data %>%
  dplyr::select(KEGG_Pathway, X.query) %>%
  tidyr::separate_rows(KEGG_Pathway, sep = ",") %>%
  dplyr::filter(KEGG_Pathway != "" & KEGG_Pathway != "-" & !is.na(KEGG_Pathway)) %>%
  dplyr::filter(grepl("map", KEGG_Pathway)) %>%
  dplyr::mutate(query = trimws(X.query), KEGG_Pathway = trimws(KEGG_Pathway)) %>%
  dplyr::select(KEGG_Pathway, X.query) %>%
  as.data.frame()

# Official KEGG pathway names, used as TERM2NAME for GSEA
```

```

kegg_names <- keggList("pathway")
term2name_kegg <- data.frame(
  PathwayID = sub("path:", "", names(kegg_names)),
  PathwayName = as.character(kegg_names)
)
term2name_kegg$PathwayName <- gsub(" - Reference pathway", "", term2name_kegg$PathwayName)

# Build the ranked gene list (by log2FoldChange) required for GSEA
gsea_ranked <- full_annotation %>%
  filter(!is.na(GeneID) & !is.na(log2FoldChange)) %>%
  group_by(GeneID) %>%
  slice_max(order_by = abs(log2FoldChange), n = 1, with_ties = FALSE) %>%
  ungroup()

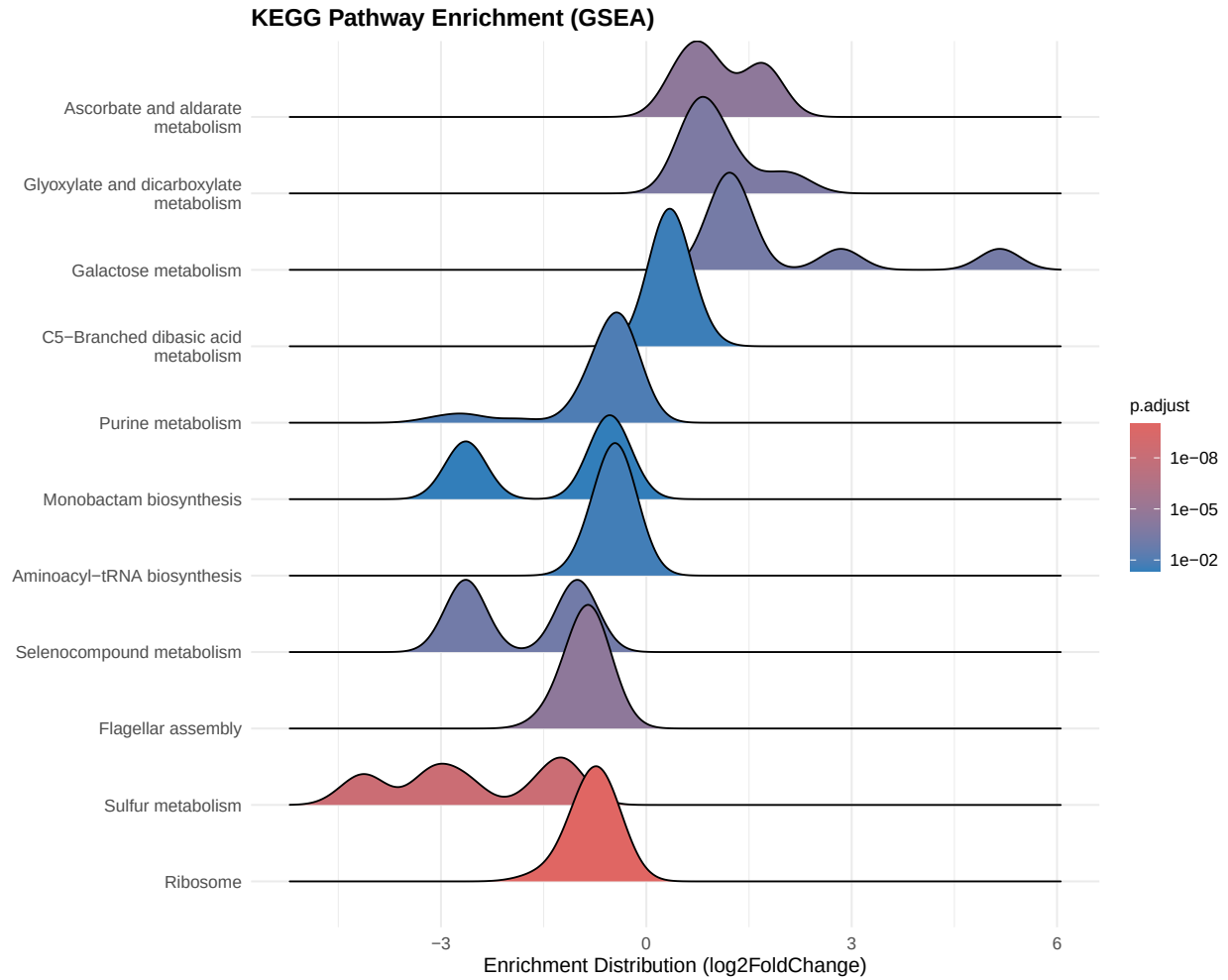
gsea_vector <- gsea_ranked$log2FoldChange
names(gsea_vector) <- gsea_ranked$GeneID
gsea_vector <- sort(gsea_vector, decreasing = TRUE)

set.seed(124)
gsea_kegg <- GSEA(geneList = gsea_vector,
  TERM2GENE = term2gene_kegg,
  TERM2NAME = term2name_kegg,
  pvalueCutoff = 0.05,
  pAdjustMethod = "BH",
  minGSSize = 10,
  maxGSSize = 500,
  eps = 0,
  seed = TRUE) # reproducible permutations (uses set.seed above)

# Drop any "Photosynthesis" pathways (case-insensitive) before plotting
gsea_kegg_filtered <- gsea_kegg
gsea_kegg_filtered@result <- gsea_kegg_filtered@result %>%
  dplyr::filter(!str_detect(tolower(Description), "photosynthesis"))

p_kegg_ridge <- ridgeplot(gsea_kegg_filtered, showCategory = 15) +
  labs(title = "KEGG Pathway Enrichment (GSEA)",
    x = "Enrichment Distribution (log2FoldChange)") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.y = element_text(size = 10),
    axis.text.x = element_text(size = 10),
    legend.title = element_text(size = 10))
p_kegg_ridge

```



```
ggsave(filename = file.path(paper_dir, "Fig2_KEGG_Enrichment_Ridgeplot.tiff"),
        plot = p_kegg_ridge, width = 10, height = 8, dpi = DPI, compression = "lzw")
```

4. Figure 3 — Protein-Protein Interaction (PPI) Network

```
deg_network <- full_annotation %>%
  filter(padj < 0.05) %>%
  dplyr::select(GeneID, Preferred_name, seed_ortholog, log2FoldChange) %>%
  mutate(search_name = coalesce(Preferred_name, GeneID)) %>%
  group_by(search_name) %>%
  slice_max(order_by = abs(log2FoldChange), n = 1, with_ties = FALSE) %>%
  ungroup()

valid_orthologs <- deg_network$seed_ortholog[!is.na(deg_network$seed_ortholog) &
                                              deg_network$seed_ortholog != ""]
tax_id <- as.numeric(strsplit(valid_orthologs[1], "\\.")[[1]][1])
```

```
string_db <- STRINGdb$new(version = "11.5", species = tax_id, score_threshold = 700)
mapped_genes <- string_db$map(as.data.frame(deg_network), "search_name", removeUnmappedRows = TRUE)
```

```
## Warning: we couldn't map to STRING 64% of your identifiers
```

```
interactions <- string_db$get_interactions(mapped_genes$STRING_id)

ppi_graph <- graph_from_data_frame(d = interactions, directed = FALSE)
ppi_graph <- simplify(ppi_graph)

if (vcount(ppi_graph) > 0) {
  # Louvain is stochastic: without a seed, module membership (and therefore the
  # per-module hub selection below) varies between runs. Seed 124 reproduces the
  # hub set shown in the published figure; 13 of the 15 hubs are stable across seeds.
  set.seed(124)
  communities <- cluster_louvain(ppi_graph)
  ppi_graph <- set_vertex_attr(ppi_graph, "cluster", value = as.numeric(membership(communities)))
  ppi_graph <- set_vertex_attr(ppi_graph, "degree", value = degree(ppi_graph))
  ppi_graph <- set_vertex_attr(ppi_graph, "betweenness", value = betweenness(ppi_graph))

  node_metadata <- data.frame(STRING_id = V(ppi_graph)$name) %>%
    left_join(mapped_genes, by = "STRING_id") %>%
    distinct(STRING_id, .keep_all = TRUE)

  ppi_graph <- set_vertex_attr(ppi_graph, "label", value = node_metadata$search_name)
  ppi_graph <- set_vertex_attr(ppi_graph, "log2FC", value = node_metadata$log2FoldChange)

  top_hubs <- node_metadata %>%
    mutate(Degree = V(ppi_graph)$degree,
           Betweenness = V(ppi_graph)$betweenness,
           Cluster = V(ppi_graph)$cluster) %>%
    group_by(Cluster) %>%
    slice_max(order_by = Betweenness, n = 3, with_ties = FALSE) %>%
    ungroup() %>%
    arrange(desc(Betweenness)) %>%
    head(15)
} else {
  message("STRINGdb could not find any interactions above the 700 threshold.")
}
```

```
if (vcount(ppi_graph) > 0) {
  net_components <- components(ppi_graph)
  if (max(net_components$csizes) > 1) {
    main_component <- induced_subgraph(
      ppi_graph, vids = which(net_components$membership == which.max(net_components$csizes)))

    tidy_net <- as_tbl_graph(main_component) %>%
      mutate(
        is_hub = name %in% top_hubs$STRING_id,
        hub_label = ifelse(is_hub, label, NA),
        clean_betweenness = replace_na(betweenness, 0),
        max_bet = max(clean_betweenness, na.rm = TRUE),
        node_size = ifelse(max_bet > 0, (clean_betweenness / max_bet) * 10 + 3, 5))
  }
}
```

```

)

p_net <- ggraph(tidy_net, layout = "kk") +
  geom_edge_link(color = "grey85", width = 0.5, alpha = 0.6) +
  geom_node_point(aes(size = node_size, color = as.factor(cluster), alpha = is_hub)) +
  scale_alpha_manual(values = c("TRUE" = 1, "FALSE" = 0.4), guide = "none") +
  geom_node_text(aes(label = hub_label),
    repel = TRUE, fontface = "bold", size = 5,
    box.padding = 1, point.padding = 0.5, max.overlaps = Inf,
    bg.color = "white", bg.r = 0.15,
    segment.color = "grey40", segment.size = 0.6) +
  # --- Color legend: Louvain clusters are shown as visual grouping only,
  # not as biologically meaningful modules. Title reflects this so the
  # reader doesn't over-interpret the color as functional annotation. ---
  scale_color_viridis_d(option = "viridis", name = "Network module\n(visual grouping)",
    labels = NULL, guide = guide_legend(
      override.aes = list(size = 5),
      label = FALSE,
      keywidth = 0.6)) +
  # --- Size legend: betweenness centrality, shown as real continuous values ---
  scale_size_continuous(name = "Betweenness\ncentrality",
    range = c(1.5, 11),
    breaks = scales::breaks_pretty(n = 4)) +
  labs(title = "Protein-Protein Interaction Network",
    subtitle = "Top 15 Bottleneck Hubs Highlighted (High Confidence STRING Score > 700)",
    # --- Clarifying caption: the analytical focus of this figure is
    # bottleneck identification (betweenness), not the cluster
    # colors, which only separate the network visually. ---
    caption = "Node color denotes Louvain clusters used for visual separation only\nand does not")
  theme_void() +
  theme(legend.position = "right",
    legend.title = element_text(face = "bold", size = 11),
    legend.text = element_text(size = 9),
    legend.spacing = unit(0.6, "cm"),
    plot.title = element_text(face = "bold", size = 18, hjust = 0.5),
    plot.subtitle = element_text(size = 14, hjust = 0.5, color = "grey40"),
    plot.caption = element_text(size = 9, color = "grey40", hjust = 0.5,
      margin = margin(t = 10)),
    plot.margin = margin(10, 10, 10, 10))

# Render once as a high-res PDF, then rasterize to TIFF at 1200 dpi
pdf_path <- file.path(paper_dir, "Fig3_PPI_Network_tmp.pdf")
tiff_path <- file.path(paper_dir, "Fig3_PPI_Network.tiff")

pdf(pdf_path, width = 12, height = 12)
print(p_net)
dev.off()

image_write(
  image_read_pdf(pdf_path, density = DPI),
  path = tiff_path, format = "tiff", compression = "lzw"
)
file.remove(pdf_path)

```

```
## PDF error: No display font for 'Symbol'
## PDF error: No display font for 'ArialUnicode'
```



5. Figure 4 — Volcano Plot: Virulence Genes Highlighted

```

# Functional keywords are searched in the eggNOG description; the gene-family
# prefixes (pel, peh, out, kdg) are matched only at the start of the gene name,
# so that e.g. "out" does not capture every "outer membrane" protein.
virulence_desc_kw <- "pectate lyase|cellulase|protease|secretion system"
virulence_name_kw <- "^ (pel|peh|out|kdg)"

# --- Relaxed significance cutoffs for virulence highlighting ---
# Original cutoffs (padj < 0.05, |log2FC| > 1 / 2-fold) were too strict and
# left almost all virulence-associated genes in the "NS" (not significant)
# bucket even when they showed meaningful shifts. Relaxed to the standard
# 1.5-fold change threshold (log2(1.5) 0.585) and a more permissive
# padj < 0.1, so more biologically relevant virulence genes are highlighted.
VIR_FC_CUTOFF <- 0.585
VIR_PADJ_CUTOFF <- 0.1

virulence_volcano_data <- full_annotation %>%
  mutate(
    Is_Virulence = if_else(
      str_detect(tolower(Description), virulence_desc_kw) |
      str_detect(tolower(Preferred_name), virulence_name_kw), "Yes", "No"),
    Category = case_when(
      Is_Virulence == "Yes" & !is.na(padj) & padj < VIR_PADJ_CUTOFF & log2FoldChange > VIR_FC_CUTOFF ~
      Is_Virulence == "Yes" & !is.na(padj) & padj < VIR_PADJ_CUTOFF & log2FoldChange < -VIR_FC_CUTOFF ~
      Is_Virulence == "Yes"
      TRUE
    ),
    Label = if_else(
      Is_Virulence == "Yes" & !is.na(padj) & padj < VIR_PADJ_CUTOFF & abs(log2FoldChange) > VIR_FC_CUTOFF,
      coalesce(
        na_if(Preferred_name, "-"),
        str_extract(GeneID, "WP_[0-9]+\\.[0-9]+"),
        GeneID),
      NA_character_)
  )

vir_colors <- c("Background" = "#BDBDBD",
               "Virulence (NS)" = "#A6D96A",
               "Virulence ↑ in Treatment" = "#D73027",
               "Virulence ↓ in Treatment" = "#4575B4")
vir_sizes <- c("Background" = 1.2, "Virulence (NS)" = 2.5,
               "Virulence ↑ in Treatment" = 3.5, "Virulence ↓ in Treatment" = 3.5)
vir_alpha <- c("Background" = 0.35, "Virulence (NS)" = 0.65,
               "Virulence ↑ in Treatment" = 1, "Virulence ↓ in Treatment" = 1)

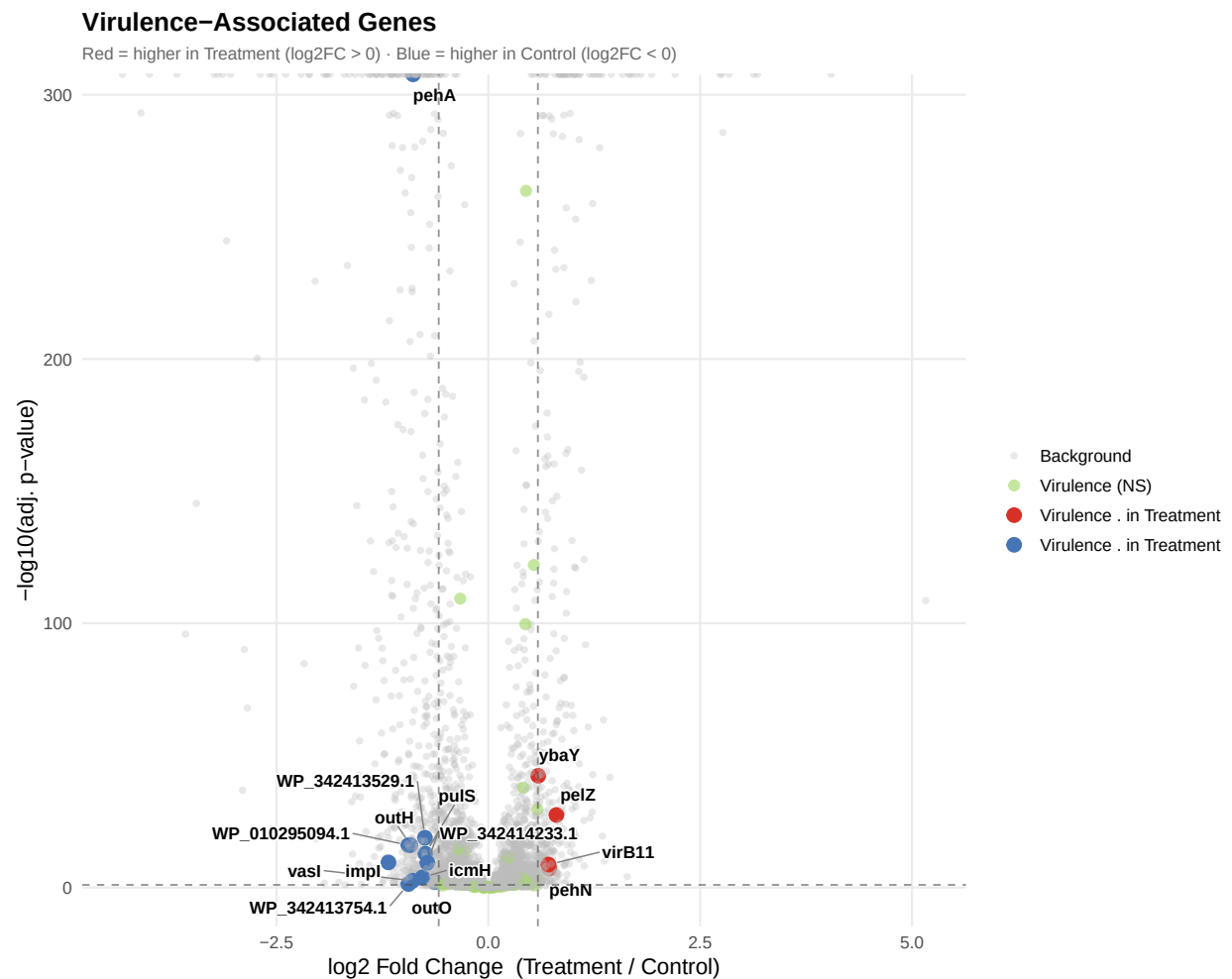
p_vir_volcano <- ggplot(virulence_volcano_data, aes(x = log2FoldChange, y = -log10(padj))) +
  geom_point(aes(color = Category, size = Category, alpha = Category)) +
  scale_color_manual(values = vir_colors, name = NULL) +
  scale_size_manual(values = vir_sizes, name = NULL) +
  scale_alpha_manual(values = vir_alpha, name = NULL) +
  geom_text_repel(data = ~ filter(.x, !is.na(Label)),
                  aes(label = Label),

```

```

size = 3.5, fontface = "bold",
box.padding = 0.5, point.padding = 0.3,
max.overlaps = Inf,
color = "black", bg.color = "white", bg.r = 0.12,
segment.color = "grey50", segment.size = 0.4) +
geom_vline(xintercept = c(-VIR_FC_CUTOFF, VIR_FC_CUTOFF), linetype = "dashed",
color = "grey30", alpha = 0.6, linewidth = 0.5) +
geom_hline(yintercept = -log10(VIR_PADJ_CUTOFF), linetype = "dashed",
color = "grey30", alpha = 0.6, linewidth = 0.5) +
labs(title = "Virulence-Associated Genes",
subtitle = "Red = higher in Treatment (log2FC > 0) · Blue = higher in Control (log2FC < 0)",
x = "log Fold Change (Treatment / Control)",
y = "-log (adj. p-value)") +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", size = 15),
plot.subtitle = element_text(size = 10, color = "grey40"),
panel.grid.minor = element_blank(),
legend.position = "right")
p_vir_volcano

```



```
ggsave(filename = file.path(paper_dir, "Fig4_Volcano_VirulenceGenes.tiff"),
        plot = p_vir_volcano, width = 10, height = 8, dpi = DPI, compression = "lzw")
```

6. Figure 5 — Heatmap: Virulence Genes

```
rownames(normcounts) <- str_extract(rownames(normcounts), "WP_[0-9]+\.[0-9]+")
normcounts_clean <- normcounts[!is.na(rownames(normcounts)), ]

virulence_matrix <- normcounts_clean[virulence_genes, ]
rownames(virulence_matrix) <- ifelse(!is.na(virulence_df$Preferred_name),
                                     virulence_df$Preferred_name,
                                     virulence_df$GeneID)

p_virulence_hm <- pheatmap(virulence_matrix,
                           scale = "row",
                           color = viridis(100),
                           treeheight_row = 15,
                           treeheight_col = 15,
                           main = "Pectobacterium Virulence Factors Expression",
                           fontsize_row = 8)
```



```
ggsave(filename = file.path(paper_dir, "Fig5_Heatmap_VirulenceGenes.tiff"),
        plot = p_virulence_hm, width = 8, height = 10, dpi = DPI, compression = "lzw")
```

7. Figure 6 — Volcano Plot: Transcriptional Regulators

```
tf_keywords <- "transcription factor|regulator|dna-binding" # matched against lower-cased text

regulator_volcano_data <- full_annotation %>%
  mutate(
    Is_Regulator = if_else(
      str_detect(tolower(Description), tf_keywords) |
      str_detect(tolower(Preferred_name), tf_keywords), "Yes", "No"),
    Category = case_when(
      Is_Regulator == "Yes" & !is.na(padj) & padj < 0.05 & log2FoldChange > 1 ~ "Regulator ↑ in Treatment",
      Is_Regulator == "Yes" & !is.na(padj) & padj < 0.05 & log2FoldChange < -1 ~ "Regulator ↓ in Treatment",
      Is_Regulator == "Yes" ~ "Regulator (NS)",
      TRUE ~ "Background"
    ),
    Label = if_else(
      Is_Regulator == "Yes" & !is.na(padj) & padj < 0.05 & abs(log2FoldChange) > 1,
      coalesce(
        na_if(Preferred_name, "-"),
        str_extract(GeneID, "WP_[0-9]+\\. [0-9]+"),
        GeneID),
      NA_character_)
  )

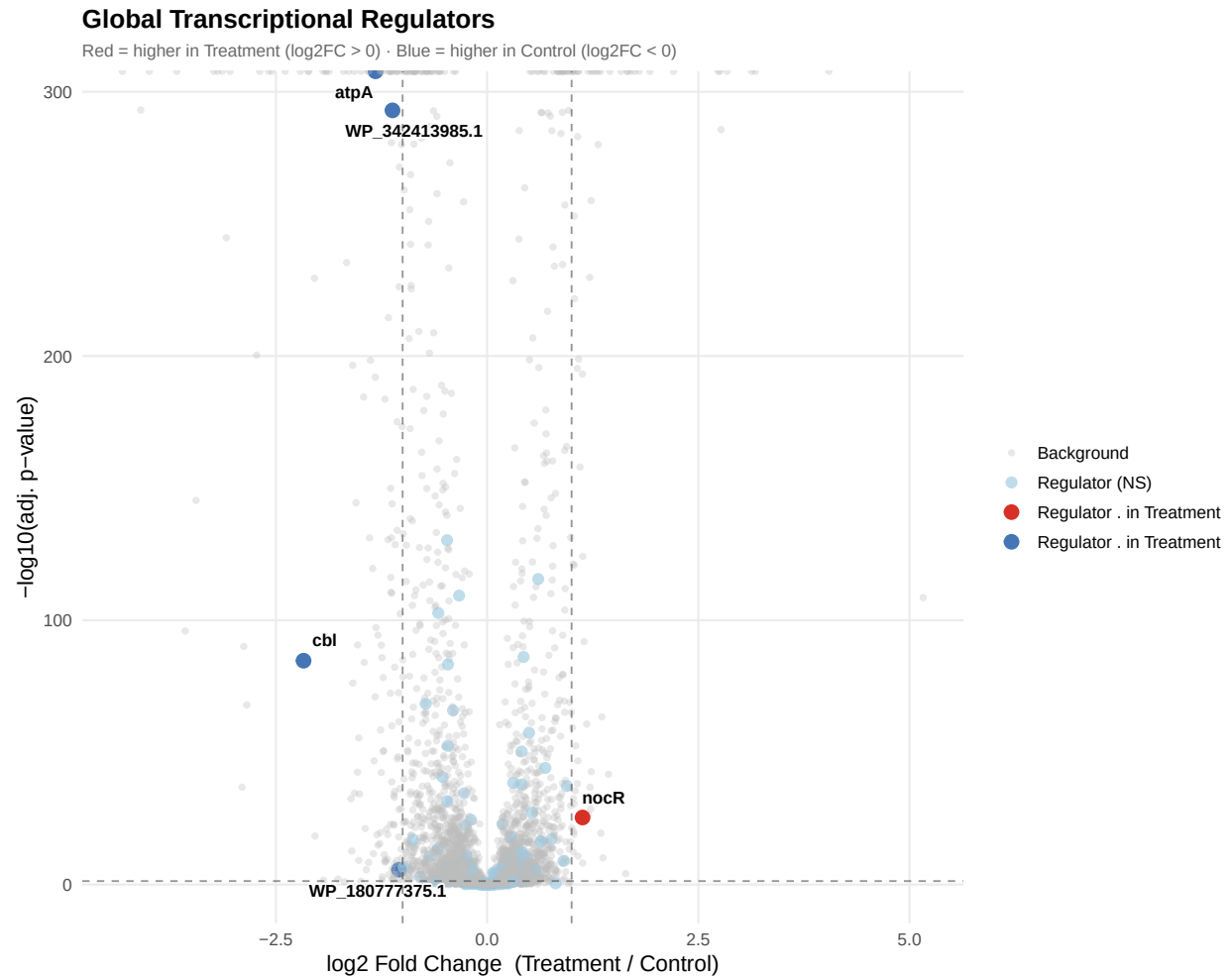
reg_colors <- c("Background" = "#BDBDBD",
               "Regulator (NS)" = "#9ECAE1",
               "Regulator ↑ in Treatment" = "#D73027",
               "Regulator ↓ in Treatment" = "#4575B4")
reg_sizes <- c("Background" = 1.2, "Regulator (NS)" = 2.5,
               "Regulator ↑ in Treatment" = 3.5, "Regulator ↓ in Treatment" = 3.5)
reg_alpha <- c("Background" = 0.35, "Regulator (NS)" = 0.65,
               "Regulator ↑ in Treatment" = 1, "Regulator ↓ in Treatment" = 1)

p_tf_volcano <- ggplot(regulator_volcano_data, aes(x = log2FoldChange, y = -log10(padj))) +
  geom_point(aes(color = Category, size = Category, alpha = Category)) +
  scale_color_manual(values = reg_colors, name = NULL) +
  scale_size_manual(values = reg_sizes, name = NULL) +
  scale_alpha_manual(values = reg_alpha, name = NULL) +
  geom_text_repel(data = ~ filter(.x, !is.na(Label)),
                 aes(label = Label),
                 size = 3.5, fontface = "bold",
                 box.padding = 0.5, point.padding = 0.3,
                 max.overlaps = Inf,
                 color = "black", bg.color = "white", bg.r = 0.12,
                 segment.color = "grey50", segment.size = 0.4) +
  geom_vline(xintercept = c(-1, 1), linetype = "dashed",
             color = "grey30", alpha = 0.6, linewidth = 0.5) +
  geom_hline(yintercept = -log10(0.05), linetype = "dashed",
             color = "grey30", alpha = 0.6, linewidth = 0.5) +
  labs(title = "Global Transcriptional Regulators",
       subtitle = "Red = higher in Treatment (log2FC > 0) · Blue = higher in Control (log2FC < 0)",
       x = "log Fold Change (Treatment / Control)",
       y = "-log (adj. p-value)") +
```

```

theme_minimal(base_size = 13) +
theme(plot.title       = element_text(face = "bold", size = 15),
      plot.subtitle    = element_text(size = 10, color = "grey40"),
      panel.grid.minor = element_blank(),
      legend.position  = "right")
p_tf_volcano

```



```

ggsave(filename = file.path(paper_dir, "Fig6_Volcano_Regulators.tiff"),
        plot = p_tf_volcano, width = 10, height = 8, dpi = DPI, compression = "lzw")

```

Session information

```
sessionInfo()
```

```

## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64

```

```

## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Mexico.utf8  LC_CTYPE=Spanish_Mexico.utf8
## [3] LC_MONETARY=Spanish_Mexico.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Mexico.utf8
##
## time zone: America/Mexico_City
## tzcode source: internal
##
## attached base packages:
## [1] stats4      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] pdftools_3.9.0          tidygraph_1.3.1
## [3] ggraph_2.2.2           igraph_2.2.2
## [5] KEGGREST_1.50.0        magick_2.9.1
## [7] STRINGdb_2.22.0       UpSetR_1.4.1
## [9] GO.db_3.22.0           AnnotationDbi_1.72.0
## [11] pathview_1.50.0        enrichplot_1.30.5
## [13] clusterProfiler_4.18.4 viridis_0.6.5
## [15] viridisLite_0.4.3      pheatmap_1.0.13
## [17] cowplot_1.2.0          lubridate_1.9.5
## [19] forcats_1.0.1          stringr_1.6.0
## [21] dplyr_1.2.0            purrr_1.2.1
## [23] readr_2.2.0            tidyr_1.3.2
## [25] tibble_3.3.1           tidyverse_2.0.0
## [27] EnhancedVolcano_1.28.2 ggrepel_0.9.7
## [29] ggplot2_4.0.3          DESeq2_1.50.2
## [31] SummarizedExperiment_1.40.0 Biobase_2.70.0
## [33] MatrixGenerics_1.22.0  matrixStats_1.5.0
## [35] GenomicRanges_1.62.1   Seqinfo_1.0.0
## [37] IRanges_2.44.0         S4Vectors_0.48.0
## [39] BiocGenerics_0.56.0    generics_0.1.4
##
## loaded via a namespace (and not attached):
## [1] splines_4.5.2          bitops_1.0-9           ggplotify_0.1.3
## [4] R.oo_1.27.1            polyclip_1.10-7        graph_1.88.1
## [7] XML_3.99-0.23          lifecycle_1.0.5        lattice_0.22-7
## [10] vroom_1.7.1            MASS_7.3-65            magrittr_2.0.4
## [13] rmarkdown_2.31         plotrix_3.8-14         yaml_2.3.12
## [16] otel_0.2.0             ggtangle_0.1.2         askpass_1.2.1
## [19] DBI_1.3.0              RColorBrewer_1.1-3     abind_1.4-8
## [22] R.utils_2.13.0         RCurl_1.98-1.19        hash_2.2.6.4
## [25] yulab.utils_0.2.4      tweenr_2.0.3           rappdirs_0.3.4
## [28] gdtools_0.5.0          tidytree_0.4.8         proto_1.0.0
## [31] codetools_0.2-20       DelayedArray_0.36.1    DOSE_4.4.0
## [34] ggforce_0.5.0          tidyselect_1.2.1       aplot_0.3.1
## [37] farver_2.1.2           jsonlite_2.0.0         ggribes_0.5.7
## [40] systemfonts_1.3.2      chron_2.3-63           tools_4.5.2

```

## [43] ggnewscale_0.5.2	treeio_1.34.0	ragg_1.5.0
## [46] snow_0.4-4	Rcpp_1.1.1	glue_1.8.0
## [49] gridExtra_2.3.1	SparseArray_1.10.10	xfun_0.55
## [52] qvalue_2.42.0	withr_3.0.3	fastmap_1.2.0
## [55] caTools_1.18.4	digest_0.6.39	timechange_0.4.0
## [58] R6_2.6.1	gridGraphics_0.5-1	qpdf_1.4.1
## [61] textshaping_1.0.4	gtools_3.9.5	RSQLite_2.4.6
## [64] R.methodsS3_1.8.2	utf8_1.2.6	fontLiberation_0.1.0
## [67] data.table_1.18.2.1	graphlayouts_1.2.5	httr_1.4.8
## [70] htmlwidgets_1.6.4	S4Arrays_1.10.1	scatterpie_0.2.6
## [73] sqldf_0.4-12	pkgconfig_2.0.3	gtable_0.3.6
## [76] blob_1.3.0	S7_0.2.1	XVector_0.50.0
## [79] htmltools_0.5.9	fontBitstreamVera_0.1.1	fgsea_1.36.2
## [82] scales_1.4.0	png_0.1-9	ggfun_0.2.1
## [85] knitr_1.51	tzdb_0.5.0	reshape2_1.4.5
## [88] curl_7.0.0	nlme_3.1-168	org.Hs.eg.db_3.22.0
## [91] cachem_1.1.0	KernSmooth_2.23-26	parallel_4.5.2
## [94] pillar_1.11.1	grid_4.5.2	vctrs_0.7.1
## [97] gplots_3.3.0	tidydr_0.0.6	cluster_2.1.8.1
## [100] Rgraphviz_2.54.0	evaluate_1.0.5	KEGGgraph_1.70.0
## [103] gsubfn_0.7	tinytex_0.60	cli_3.6.5
## [106] locfit_1.5-9.12	compiler_4.5.2	rlang_1.3.0
## [109] crayon_1.5.3	labeling_0.4.3	plyr_1.8.9
## [112] fs_2.1.0	ggiraph_0.9.6	stringi_1.8.7
## [115] BiocParallel_1.44.0	Biostrings_2.78.0	lazyeval_0.2.2
## [118] GOSemSim_2.36.0	fontquiver_0.2.1	Matrix_1.7-4
## [121] hms_1.1.4	patchwork_1.3.2	bit64_4.6.0-1
## [124] memoise_2.0.1	ggtree_4.0.5	fastmatch_1.1-8
## [127] bit_4.6.0	ape_5.8-1	gson_0.2.0